山东大学 计算机科学与技术 学院

云计算技术 课程实验报告

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实验题目：负载均衡		
实验学时：2	实验日期： 2025.5.21	

实验目的：使用 Docker 部署 Nginx 和 Tomcat 实现负载均衡

具体包括：

1、要求部署一台 Nginx 和四台 Tomcat 服务器

2、Ngnix 需要实现四种策略：

1）轮询法；

2）随机法；

3）加权轮询法，权重为 4，2，1，5；

4）IP Hash。

3、需要设计一个前端网页，通过该网页来显示负载均衡的效果。例如，每次访问页面时，页面可以显示当前处理请求信息，以体现 Nginx 的负载均衡策略。

硬件环境：

PC

软件环境：

Linux 或 Windows

实验步骤与内容：

1、配置 Dockerfile 文件和四种调度策略文件，配置 docker compose 需要的.yml 文件，配置前端展示.jsp 文件

2、构建 docker 镜像如下：

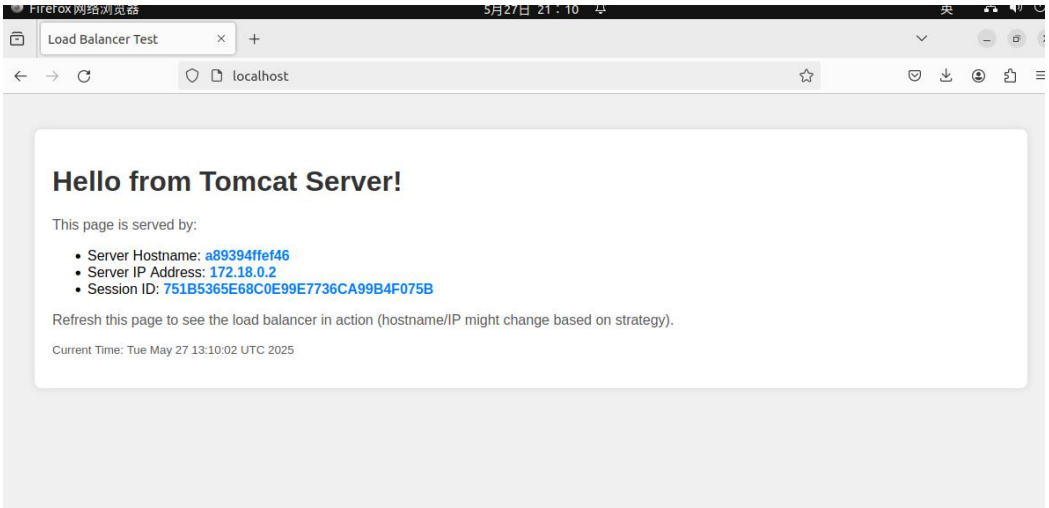
```
xuerunze@xuerunze-virtual-machine:~/Documents/cloud_calculate_ex/ex11$ docker co
mpose up --build -d
WARN[0000] /home/xuerunze/Documents/cloud_calculate_ex/ex11/docker-compose.yml:
the attribute `version` is obsolete, it will be ignored, please remove it to avo
id potential confusion
Compose can now delegate builds to bake for better performance.
 To do so, set COMPOSE_BAKE=true.
[+] Building 33.3s (13/19)                                      docker:default
 => => sha256:0622fac788edde5d30e7bbd2688893e5452a19ff 19.92MB / 29.72MB  32.9s
 => => sha256:9ea5a509df298cf38e68f05470427ca0916c9487 16.97MB / 16.97MB  29.6s
 => => sha256:5237d93da24e4debba6f2e8318814c8c3991df 145.64MB / 145.64MB  23.7s
 => => sha256:dced65dce150f2f9daeae4c3d2cab6ef042dd5972f0c1d 160B / 160B  23.8s
 => => sha256:96103e3519e13ac7a5535cc613dbcefc69ed92bd01 2.28kB / 2.28kB  24.0s
 => => sha256:a75eb051a8344c6f35881a89b445605c42d1ecb27e6c96 139B / 139B  24.1s
 => => sha256:4f4fb700ef54461cfa02571ae0db9a0dc1e0cdb5577484a6 32B / 32B  24.2s
 => => sha256:dbd15ed7db256746beb0aede8f6483b13d1dd8c8 13.93MB / 13.93MB  27.5s
 => [tomcat4 internal] load build context                          0.0s
```

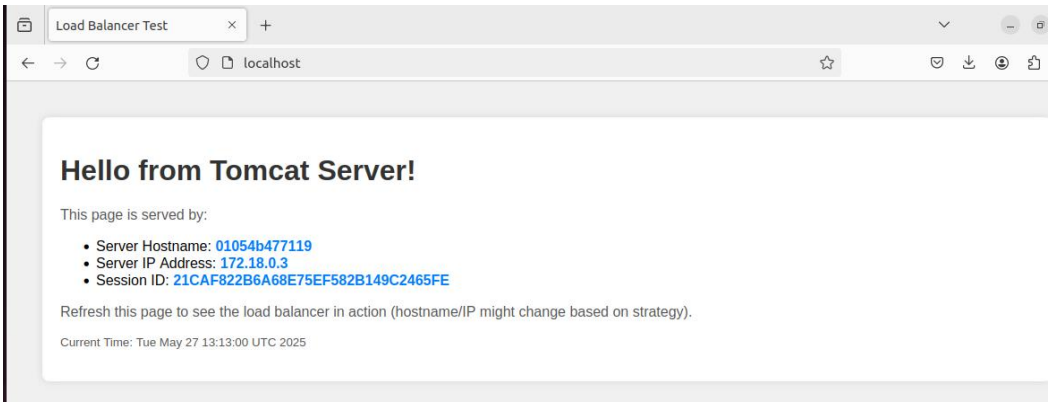
```
=> => writing image sha256:7b694c3acd8ec31d67249002ef8da58113a3ede94c634  0.0s
=> => naming to docker.io/library/ex11-tomcat3                           0.1s
=> [tomcat2] resolving provenance for metadata file                      0.6s
=> [tomcat1] resolving provenance for metadata file                      0.6s
=> [tomcat4] resolving provenance for metadata file                      0.6s
=> [tomcat3] resolving provenance for metadata file                      0.6s
+] Running 10/10
✔ tomcat1                Built                                           0.0s
✔ tomcat2                Built                                           0.0s
✔ tomcat3                Built                                           0.0s
✔ tomcat4                Built                                           0.0s
✔ Network ex11_app-net   Created                                         0.3s
✔ Container tomcat04     Started                                         7.0s
✔ Container tomcat01     Started                                         7.1s
✔ Container tomcat02     Started                                         7.1s
✔ Container tomcat03     Started                                         7.2s
✔ Container nginx_lb     Started                                         9.0s
```

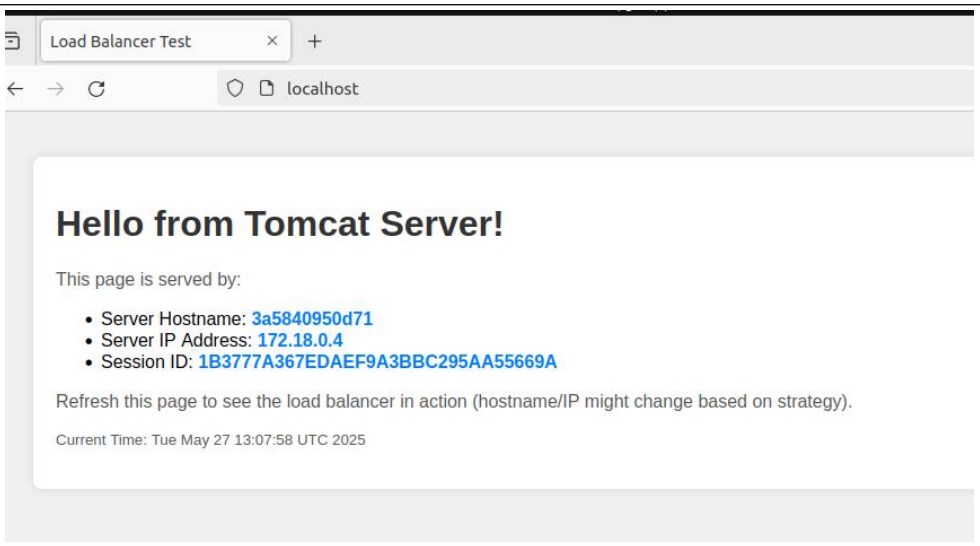
3、每次更换负载均衡策略后，重新构建镜像

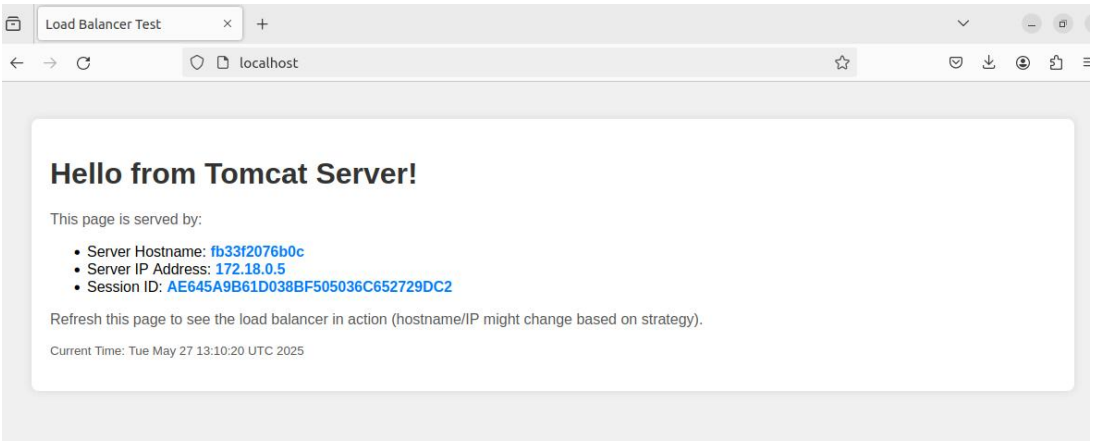
结论分析与体会：

1、使用轮询法结果如下：刷新页面重新提交访问后，Server IP 地址会在 172.18.02-172.18.05 之间轮询替换。

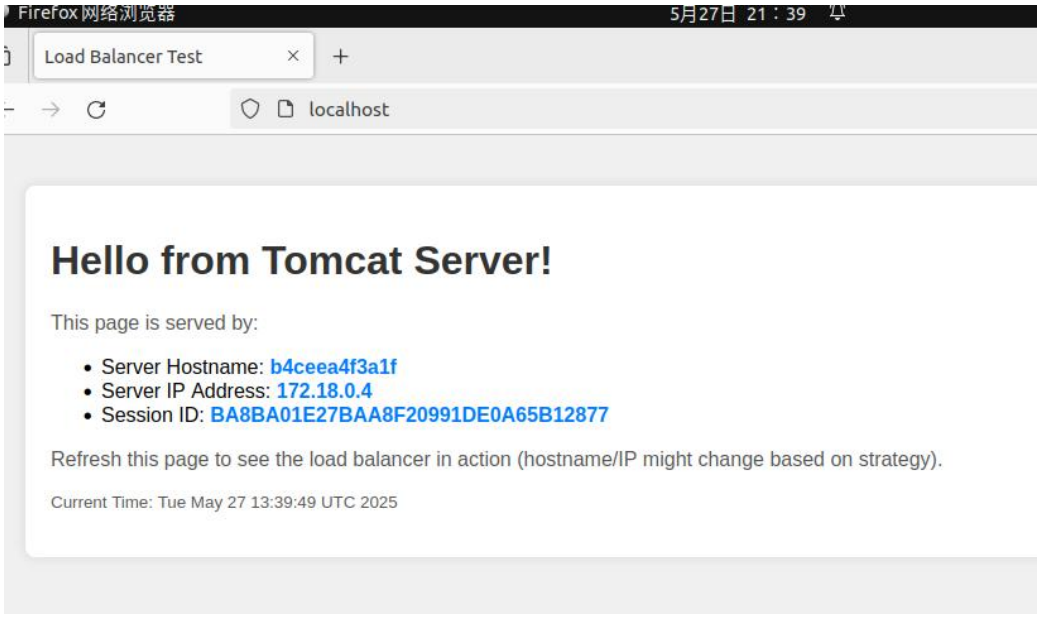


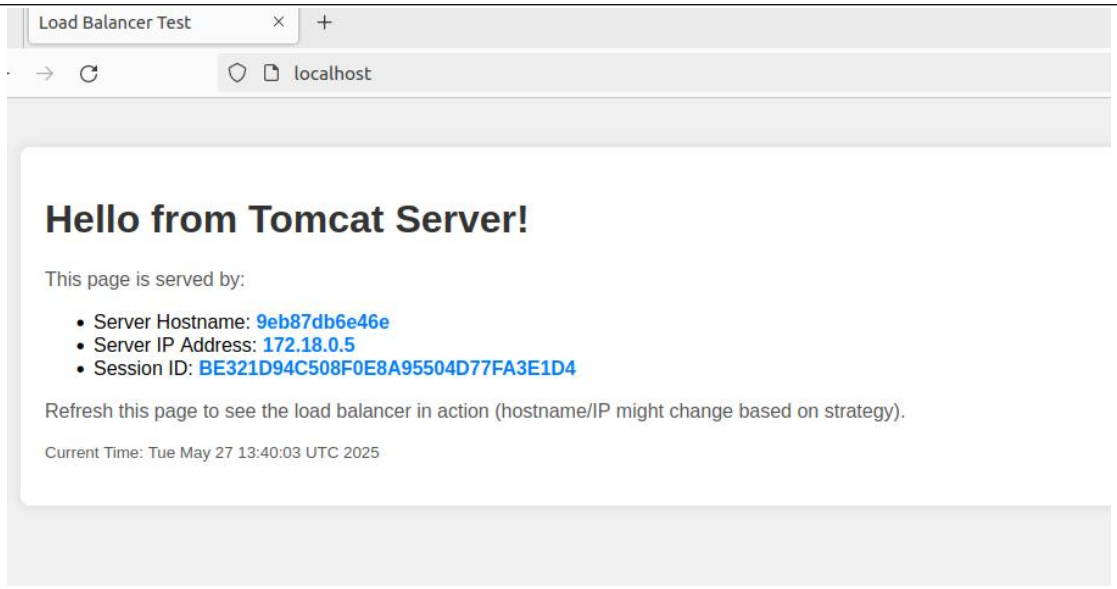


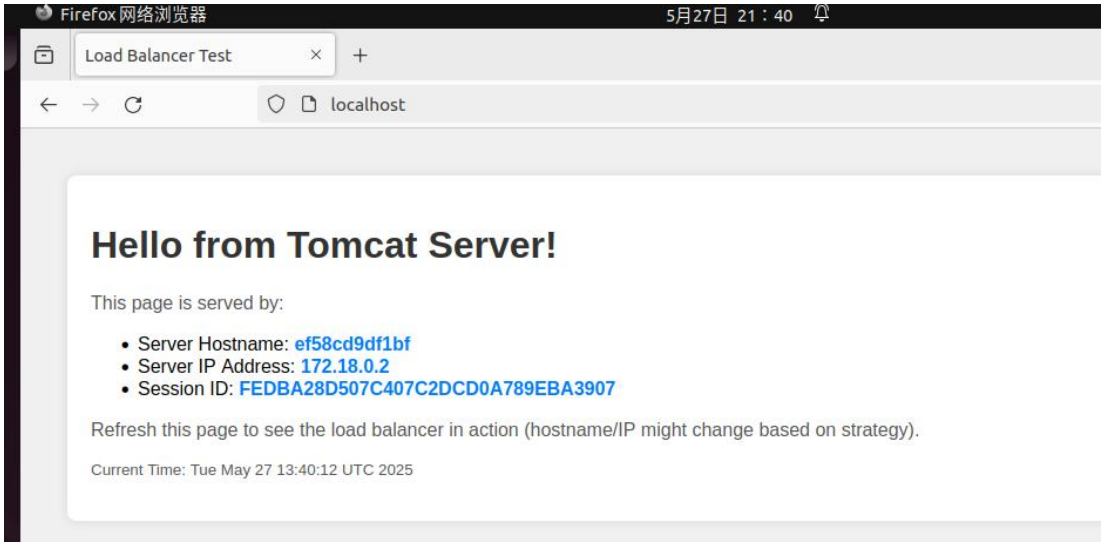


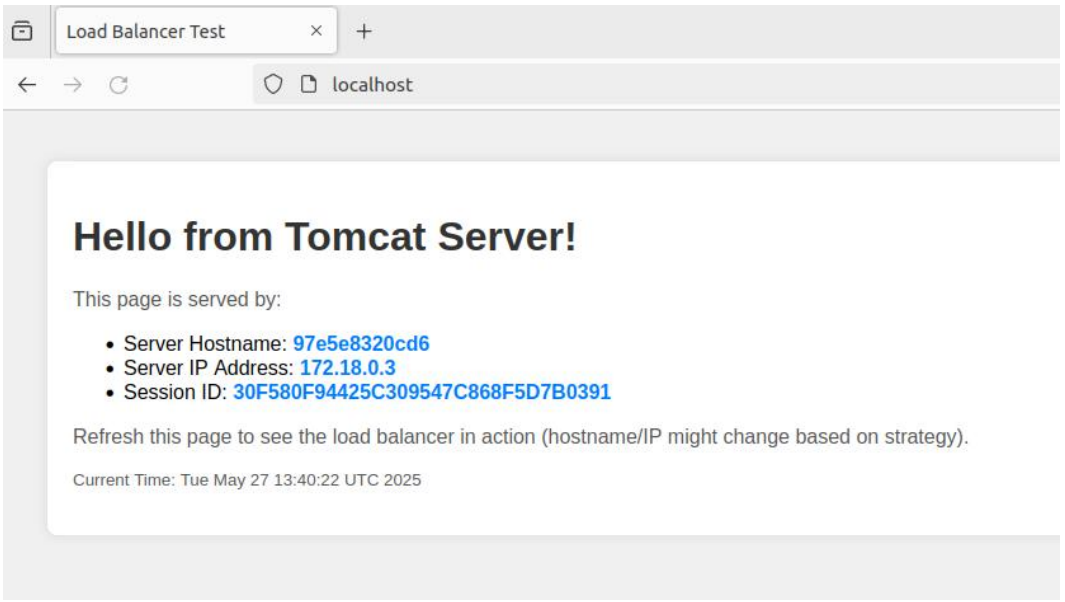


2、使用随机法（因为随机法的目的是尽量使所有服务器均匀负载，所以使用最小访问来代替随机，选择最少次数访问的服务器作为这次选定的服务器）




Load Balancer Test
localhost
Hello from Tomcat Server!
This page is served by:
Server Hostname: 9eb87db6e46e
Server IP Address: 172.18.0.5
Session ID: BE321D94C508F0E8A95504D77FA3E1D4
Refresh this page to see the load balancer in action (hostname/IP might change based on strategy).
Current Time: Tue May 27 13:40:03 UTC 2025

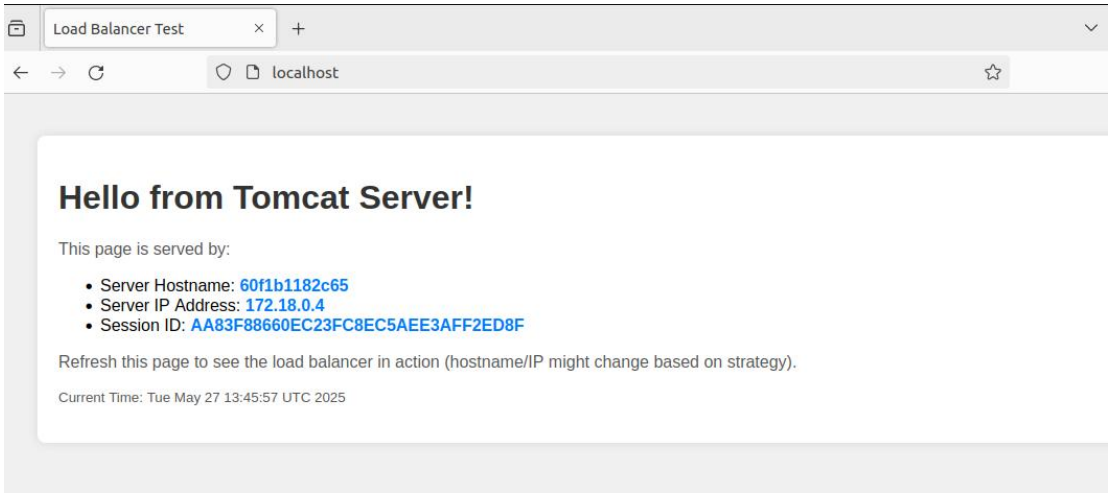

Firefox 网络浏览器
5月27日 21：40
Load Balancer Test
localhost
Hello from Tomcat Server!
This page is served by:
Server Hostname: ef58cd9df1bf
Server IP Address: 172.18.0.2
Session ID: FEDBA28D507C407C2DCD0A789EBA3907
Refresh this page to see the load balancer in action (hostname/IP might change based on strategy).
Current Time: Tue May 27 13:40:12 UTC 2025

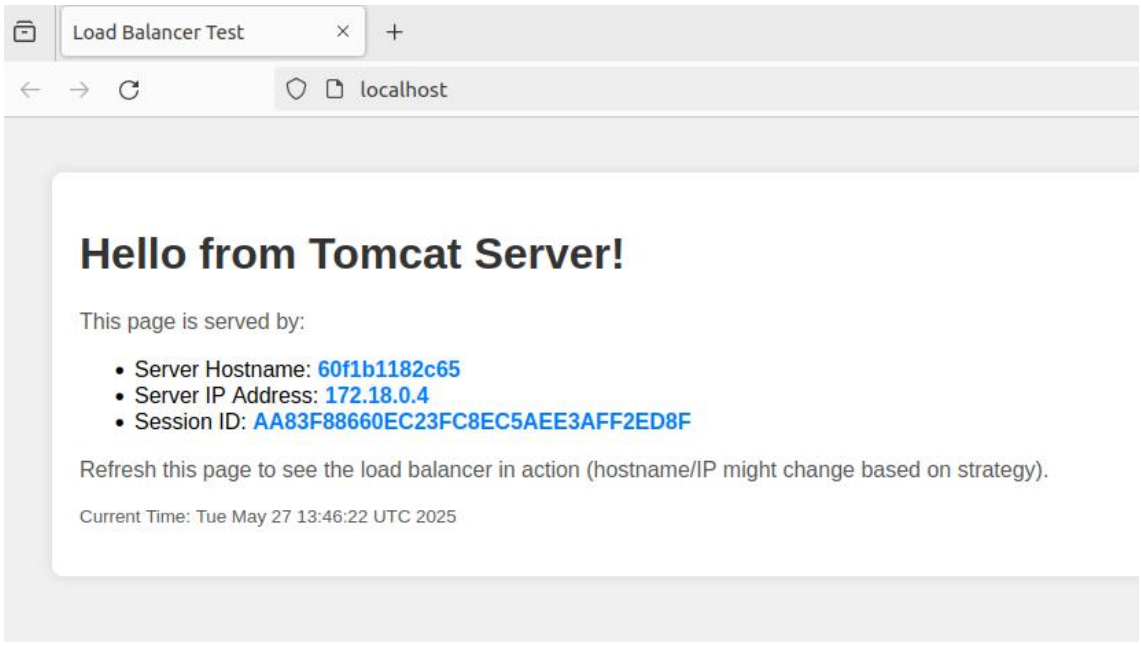

Load Balancer Test
localhost
Hello from Tomcat Server!
This page is served by:
Server Hostname: 97e5e8320cd6
Server IP Address: 172.18.0.3
Session ID: 30F580F94425C309547C868F5D7B0391
Refresh this page to see the load balancer in action (hostname/IP might change based on strategy).
Current Time: Tue May 27 13:40:22 UTC 2025

3、加权法（权重比例为 4，2，1，5）

进行 12 次访问。统计负载情况，2 号服务器访问 2 次，3 号服务器访问 5 次，4 号服务器访问 1 次，5 号服务器访问 4 次。

4、哈希法，因为 IP 地址不变，所以哈希法访问的服务器是固定的，在本次实验中，访问服务器为 4 号服务器。





代码附录：

docker-compose.yml

```
version: '3.8'

services:
  tomcat1:
```

```
    build: ./tomcat_app
    container_name: tomcat01
    networks:
      - app-net

  tomcat2:
    build: ./tomcat_app
    container_name: tomcat02
    networks:
      - app-net

  tomcat3:
    build: ./tomcat_app
    container_name: tomcat03
    networks:
      - app-net

  tomcat4:
    build: ./tomcat_app
    container_name: tomcat04
    networks:
      - app-net

  nginx:
    image: nginx:latest
    container_name: nginx_lb
    ports:
      - "80:80"
    volumes:
      - ./nginx.conf:/etc/nginx/nginx.conf:ro
    depends_on:
      - tomcat1
      - tomcat2
      - tomcat3
      - tomcat4
    networks:
      - app-net

networks:
  app-net:
    driver: bridge
```

Dockerfile

```
FROM tomcat:9-jdk11-temurin
RUN rm -rf /usr/local/tomcat/webapps/ROOT
COPY ./webapp/ROOT /usr/local/tomcat/webapps/ROOT/
EXPOSE 8080
```

Index.jsp

```
<%@ page import="java.net.InetAddress" %>
<%@ page contentType="text/html;charset=UTF-8" language="java" %>
<html>
<head>
    <title>Load Balancer Test</title>
    <style>
        body { font-family: Arial, sans-serif; margin: 40px; background-color: #f0f0f0; }
        .container { background-color: #fff; padding: 20px; border-radius: 8px; box-shadow: 0 0 10px rgba(0,0,0,0.1); }
        h1 { color: #333; }
        p { color: #555; }
        strong { color: #007bff; }
    </style>
</head>
<body>
    <div class="container">
        <h1>Hello from Tomcat Server!</h1>
        <p>This page is served by:</p>
        <ul>
            <li>Server Hostname: <strong><%= InetAddress.getLocalHost().getHostName() %></strong></li>
            <li>Server IP Address: <strong><%= InetAddress.getLocalHost().getHostAddress() %></strong></li>
            <li>Session ID: <strong><%= session.getId() %></strong></li>
        </ul>
        <p>Refresh this page to see the load balancer in action (hostname/IP might change based on strategy).</p>
        <p><small>Current Time: <%= new java.util.Date() %></small></p>
    </div>
</body>
</html>
```

nginx_ip_hash.conf

```
worker_processes auto;
```

```
events { worker_connections 1024; }
http {
    upstream tomcat_backend {
        ip_hash;
        server tomcat1:8080;
        server tomcat2:8080;
        server tomcat3:8080;
        server tomcat4:8080;
    }
    server {
        listen 80; server_name localhost;
        location / {
            proxy_pass http://tomcat_backend;
            proxy_set_header Host $host; proxy_set_header X-Real-IP $remote_addr;
            proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for; proxy_set_header X-Forwarded-Proto $scheme;
        }
    }
}
```

nginx_random.conf

```
worker_processes auto;
events { worker_connections 1024; }
http {
    upstream tomcat_backend {
        least_conn;
        server tomcat1:8080;
        server tomcat2:8080;
        server tomcat3:8080;
        server tomcat4:8080;
    }
    server {
        listen 80; server_name localhost;
        location / {
            proxy_pass http://tomcat_backend;
            proxy_set_header Host $host; proxy_set_header X-Real-IP $remote_addr;
            proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for; proxy_set_header X-Forwarded-Proto $scheme;
        }
    }
```

}

nginx_round_robin.conf

```
worker_processes auto;
events { worker_connections 1024; }
http {
    upstream tomcat_backend {
        server tomcat1:8080;
        server tomcat2:8080;
        server tomcat3:8080;
        server tomcat4:8080;
    }
    server {
        listen 80; server_name localhost;
        location / {
            proxy_pass http://tomcat_backend;
            proxy_set_header Host $host; proxy_set_header X-Real-IP $remote_addr;
            proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for; proxy_set_header X-Forwarded-Proto $scheme;
        }
    }
}
```

nginx_weighted_round_robin.conf

```
worker_processes auto;
events { worker_connections 1024; }
http {
    upstream tomcat_backend {
        server tomcat1:8080 weight=4;
        server tomcat2:8080 weight=2;
        server tomcat3:8080 weight=1;
        server tomcat4:8080 weight=5;
    }
    server {
        listen 80; server_name localhost;
        location / {
            proxy_pass http://tomcat_backend;
            proxy_set_header Host $host; proxy_set_header X-Real-IP $remote_addr;
```

```
                proxy_set_header    X-Forwarded-For    $proxy_add_x_forwarded_for;
proxy_set_header X-Forwarded-Proto $scheme;
            }
        }
}
```